

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



Sample Report Metrics

Reading the sample report

RR057_covid19_mngs >

RR_hela2_2-5ng_SEQ005_O10_S159 ▾

[Sample Details](#)



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Metagenomic

Metagenomics - Simplified

PROTOTYPE

Antimicrobial Resistance

Taxon name



Name Type: Scientific ▾

Background: None ▾

Categories ▾

Threshold filters ▾

Read Specificity: All ▾

1420 total rows.



>	Taxon ▾	Score	Z Score	rPM ▾	r ▾	contig ▾	contig r ▾	%id ▾	L ▾	E value ▾	NT NR
>	Alphapapillomavirus (1 viral species: ● 1)	-	-	1,391.3 1,354.8	80,766 78,650	5 4	80,122 79,364	100.0 100.0	4,949.7 652.7	10 ⁻³⁰⁴ 10 ⁻³⁰⁸	
>	Providencia (1 bacterial species)	-	-	316.7 0.0	18,385 0	1 0	18,385 0	99.8 0.0	1,354.0 0.0	10 ⁻³⁰⁸ 0	
>	Escherichia (3 bacterial species: ● 1)	-	-	46.0 43.6	2,672 2,531	32 39	1,067 1,234	99.8 96.0	717.5 207.1	10 ⁻¹⁵⁷ 10 ⁻²³⁶	
>	Delftia (20 bacterial species)	-	-	45.2 70.2	2,625 4,073	50 59	303 2,248	98.9 78.2	160.3 38.8	10 ⁻⁸⁸ 10 ⁻¹³⁷	
>	non-genus-specific reads in family Bradyrhizobiaceae (2 bacterial species)	-	-	42.2 0.1	2,451 7	0 0	0 0	99.7 96.7	1,448.0 38.3	10 ⁻³⁰⁸ 10 ⁻³⁰⁸	
>	Sphingomonas (33 bacterial species)	-	-	37.4 2.8	2,172 161	3 3	1,981 22	97.6 96.2	2,342.1 48.3	10 ⁻²⁸¹ 10 ⁻²⁴⁵	

Gaining confidence in species calls using report metrics

Metric	Definition	Interpretation
rPM	Number of reads aligning to taxon per million reads sequenced (normalized metric)	Higher rPM = more confidence in the hit
r	Number of reads aligning to taxon	Higher r = more confidence in the hit
Contig	Number of assembled contigs	Longer contig = more confidence
Contig r	Number of reads aligning to assembled contigs	Higher Contig r = more confidence
% id	Average % identity of alignment to accession	Should be taken into account with other metrics
L	Average length of alignment of contigs and reads	Longer alignment = more confidence (if it's divergent, L may not be high)
E value	Average expect value of alignment	Lower = more confidence

Each metric shows 2 results

NT = nucleotide database

NR = protein database

>	Taxon ▾	Score	Z Score	rPM ▾	r ▾	contig ▾	contig r ▾	%id ▾	L ▾	E value ▾	NT NR
▾	▢ Alphapapillomavirus (1 viral species: ● 1)	-	-	1,391.3 1,354.8	80,766 78,650	5 4	80,122 79,364	100.0 100.0	4,949.7 652.7	10 ⁻³⁰⁴ 10 ⁻³⁰⁸	
	▢ Alphapapillomavirus 7 Known Pathogen	-	-	1,391.3 1,354.8	80,766 78,650	5 4	80,122 79,364	100.0 100.0	4,949.7 652.7	10 ⁻³⁰⁴ 10 ⁻³⁰⁸	

Each metric shows 2 results

NT = nucleotide database

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>	Taxon ▾	Score	Z Score	rPM ▾	r ▾	contig ▾	contig r ▾	%id ▾	L ▾	E value ▾	NT NR
▾	Alphapapillomavirus (1 viral species: ● 1)	-	-	1,391.3 1,354.8	80,766 78,650	5 4	80,122 79,364	100.0 100.0	4,949.7 652.7	10 ⁻³⁰⁴ 10 ⁻³⁰⁸	
	Alphapapillomavirus 7 Known Pathogen	-	-	1,391.3 1,354.8	80,766 78,650	5 4	80,122 79,364	100.0 100.0	4,949.7 652.7	10 ⁻³⁰⁴ 10 ⁻³⁰⁸	

General Interpretation Notes:

- Reads aligning to both databases: Good sign that the hit is true (more to it than that)
- If a read is viral and only has a hit to the NR database: Hit could represent a divergent virus
- If a hit is bacterial and it only has a hit to the NT database: Sequences may be aligning to the 16S rRNA region, which is not in the NR databases (should not be trusted)

Filters are helpful at removing false hits, but not perfect

Using the following filters is a good place to start:

- **NT L $\geq$ 50**
 - I only want to see taxa with at least 50 bp that align to the nucleotide database in NCBI
- **NT rPM $\geq$ 10**
 - I only want to see taxa with at least 10 reads per million that align to the nucleotide database. Reads per million is a normalized value that takes into account total reads.
- **NR rPM $\geq$ 1**
 - I only want to see taxa with at least 1 read that aligns to the protein database in NCBI. Having at least 1 hit to the protein database ensures that you are not hitting a conserved region.

Read more about filtering [here](#).

The coverage visualization is valuable for evaluating hits

Alphapapillomavirus 7 Coverage

[View read-level visualization >](#)

MF288662.1 - Human papillomavirus type 18 isolate 1359486_N-P, c... ▼

10 viewable accessions (46 total) ⓘ



Reference NCBI Entry [MF288662.1 - Human p...](#)

Aligned Contigs 1

Coverage Depth 936.8x

Max Alignment Length 7857

Reference Length 7857

Aligned Loose Reads 0

Coverage Breadth 100.0%

Avg. Mismatched % 0.0%

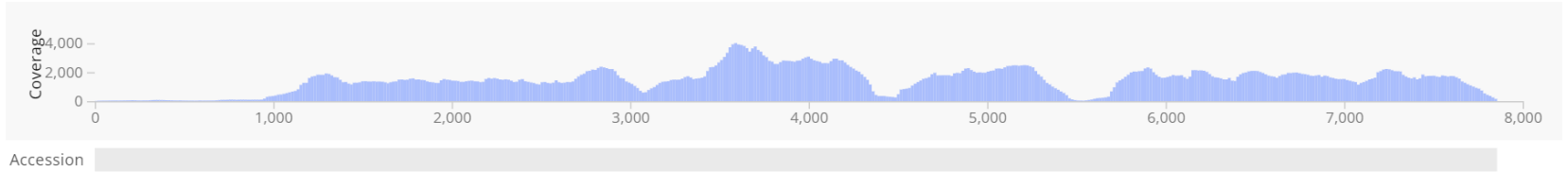


Reference Accession

Contigs (1)

Take into consideration the depth and breadth of the coverage (the higher, the better)

depth



breadth

Taking into account coverage breadth and depth

Depth	Breadth	How to Interpret
High	High	Provides you with confidence that this is a true hit!
Low	High	If there is coverage across multiple regions of the genome, this is likely a real hit. Evaluate the assembled contigs (NT L) and ensure the alignment is good (% identity).
High	Low	Depends on whether the small portion that has high coverage is unique to that particular taxon and on the quality of the alignment and assembly.
Low	Low	May be a result of genomic similarity at the 16S rRNA region (for bacteria)

What do we think about this viral hit?

Chikungunya virus Coverage

MK468610.1 - Chikungunya virus isolate CHRF_0094_11-22-2017, co... ▾

10 viewable accessions (79 total) ⓘ



NCBI Reference [MK468610.1 - Chikungunya vi...](#)

Aligned Contigs 2

Coverage Depth 3415.7x

Max Alignment Length 11818

Custom Reference

Aligned Loose Reads 99

Coverage Breadth 100.0%

Avg. Mismatched % 0.6%

Reference Length 11818



Reference Accession

NT Contigs (2)

Loose NT Reads (99)



What do we think about this bacterial hit?

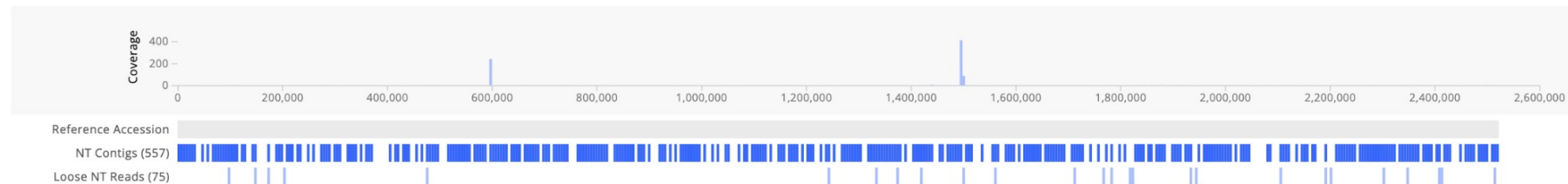
Cutibacterium acnes Coverage

CP003196.1 - Propionibacterium acnes Type1A2 P.acn17, complete ...

16 viewable accessions (32 total)



NCBI Reference	CP003196.1 - Propionibacteri...	Aligned Contigs	557	Coverage Depth	1.7x	Max Alignment Length	3227
Custom Reference		Aligned Loose Reads	75	Coverage Breadth	7.0%	Avg. Mismatched %	0.3%
Reference Length	2522885						



What do we think about this bacterial hit?

Providencia rettgeri Coverage

[View read-level visualization >](#)

CP029736.1 - *Providencia rettgeri* strain AR_0082 chromosome, co... ▼

1 viewable accessions



Reference NCBI Entry [CP029736.1 - Providenc...](#)

Aligned Contigs

1

Coverage Depth

0.4x

Max Alignment Length 1354

Reference Length 4309166

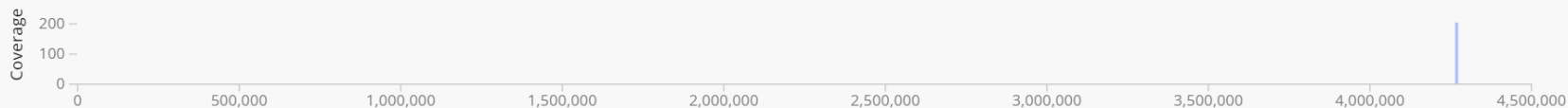
Aligned Loose Reads

0

Coverage Breadth

0.0%

Avg. Mismatched % 0.2%



Reference Accession

Contigs (1)

BLAST can be used for hit confirmation

BLASTN Contig

Escherichia

BLASTN compares nucleotide query sequence(s) to the nucleotide (NT) databases in NCBI. [Learn more](#)

BLASTN contigs helps you:

- Confirm hits.
- Determine region of coverage.
- Collect metrics for publication.

Make a selection

Up to 3 of the longest contigs are available to BLAST. The more contigs you choose, the longer the results will take to load.

☐ Contig name	Number of reads	Contig length
☐ NODE_3_length_2853_cov_9.641530	465	2901
☐ NODE_26_length_371_cov_14.582278	88	378
☐ NODE_13_length_816_cov_3.897503	68	830

ContinueCancel

Let's do our own analysis!

Current Activity: CZ ID Training Reference Tables & Worksheets_Illumina (pg 12-13)

Next Slides: DL7-I_Background models